# Architecture Ablation: Rationale and Protocol

SISA-based exact class unlearning — supporting note for the experimental section

## 1. The question the ablation answers

Is the unlearning result a property of the framework, or of the backbone?

Every result currently reported — 83.26% system accuracy, an 87.53% oracle ceiling, the routing gap, and the measured retrain fractions — comes from a single compact custom CNN. From those numbers alone a reader cannot tell whether they are properties of class-isolated sharding with class-sequential slicing, or artifacts of one small network. An architecture ablation is the only thing that separates the two.

## 2. Reasons, in order of argumentative strength

### 2.1 Backbone-independence of exactness

- If membership-inference AUC ≈ 0.5 and prediction agreement with the from-scratch reference hold at both backbones, that is evidence the guarantee follows from the algorithm — retraining from a class-clean checkpoint with scrubbed replay — rather than from a particular model. Exactness should be backbone-invariant, and demonstrating it is worth more than any accuracy gain.

### 2.2 It separates framework cost from backbone limitation

- The oracle ceiling is 87.53%. Is that the price of class-sequential slicing, or simply a weak CNN? If a residual backbone lifts the ceiling, the earlier gap was architectural. If it does not, the ceiling is structural to the framework. Either outcome is informative, and neither can be established from a single backbone.

### 2.3 Fair comparison to prior work

- ARCANE reports on WideResNet. Comparing a custom CNN against their WRN conflates architecture with method, leaving any accuracy difference uninterpretable. Reporting at a standard backbone is a precondition for an honest baseline comparison.

### 2.4 It preempts the weak-baseline criticism

- The single-CNN baseline of 81.67% is soft for CIFAR-10, and makes the sharded system look favourable partly by comparison. Results at two backbones remove that objection rather than inviting it.

## 3. Draft text for the paper

> Architecture ablation. All primary results use a compact custom CNN. To verify that the unlearning guarantees and efficiency characteristics are properties of the framework rather than of a particular backbone, we repeat the full pipeline with a CIFAR-style residual network, holding the partition, replay schedule, routing mechanism, and optimizer fixed. We report accuracy, retraining cost, and the exactness metrics for both, since accuracy and unlearning efficiency trade against one another through model size.

## 4. What the results table must contain

Both axes must appear, or the ablation misleads by showing only the accuracy side of a trade.

| backbone | params | oracle ceiling | system acc | retrain MACs | MIA-AUC | pred. agree |
|---|---|---|---|---|---|---|
| custom CNN | 0.62M | 87.53% | 83.26% | 1.0x | — | — |
| ResNet-20 | 0.27M | — | — | 3.8x | — | — |

The 3.8x retraining cost is not a weakness to bury — it is the point. It exposes the accuracy/efficiency trade explicitly, instead of reporting a single arbitrary operating point.

## 5. Why a resolution-derived backbone, not a fixed one

The backbone is derived rather than chosen, consistent with how the number of shards K, the per-shard slice count S_k, and the slice budget S_max are derived. Stages are added until the feature map reaches 8x8 before global average pooling:

```
n_stages = ceil(log2(input_size / 8)) + 1        widths = [16 * 2^i]
32x32  ->  3 stages  [16, 32, 64]        (reproduces ResNet-20 exactly)
64x64  ->  4 stages  [16, 32, 64, 128]
```

At 32x32 this reproduces ResNet-20 exactly, so the ablation is not a departure from the standard architecture — it is the standard architecture with its resolution assumption made explicit.

## 6. Measured parameter budget per shard

Class-isolated sharding gives each specialist only |C_k| * n / N_C samples, so the relevant quantity is parameters per training sample within a shard, not dataset size.

| dataset | res | K | img/shard | stages | depth | params | p/sample | current |
|---|---|---|---|---|---|---|---|---|
| CIFAR-10 | 32 | 2 | 20,250 | 3 | 20 | 272,149 | 13 | 31 |
| CIFAR-100 | 32 | 13 | 2,835 | 3 | 20 | 272,279 | 96 | 219 |
| Tiny-ImageNet | 64 | 25 | 3,600 | 4 | 26 | 1,093,848 | 304 | 173 |
| Tiny-ImageNet | 64 | 25 | 3,600 | 4 | 18 | 701,208 | 195 | 173 |

The last row uses 2 blocks per stage instead of 3. On 32x32 datasets the residual backbone more than halves parameters-per-sample against the current CNN. At 64x64 the derived fourth stage makes it worse, because depth grows with resolution while shards shrink with class count.

## 7. Limitations to state explicitly

- Two backbones spanning a 2.3x parameter range show the result is not specific to one model. They do not establish that it holds for all. Claim "consistent across two backbones", not "backbone-independent".

- BatchNorm normally degrades class-incremental learning through cross-task statistics bias. Complete balanced replay removes that failure mode here — but only while replay stays complete. State this, or a reviewer familiar with the continual-learning literature will raise it as an objection.
- The derived depth grows with input resolution while shards shrink with class count. At Tiny-ImageNet the 4-stage variant is worse on parameters-per-sample than the current CNN, so blocks per stage must itself be constrained by shard size.